



SoutheastCon 2026 Circuit Design Competition

(Design a water based fire suppression system)

The Mission

In the fictional coastal town of Port Resolute, a young U.S. startup called **Aquila Aerospace** had earned a contract to launch its first rocket, *Firebird-1*. The rocket was powerful. Too powerful.

Engineers predicted flame temperatures that could shatter the concrete and melt launch infrastructure without an aggressive deluge system. The company's solution? They needed a high-flow **DC motor-driven water pump for a water suppression system** designed to flood the launch pad at T-0 and smother reflected exhaust energy.

But Aquila's leadership had a problem.

They didn't just need a pump.

They needed **a team that could prove it was safe.**

And that team would be you – the students of the SoutheastCon 2026 Circuit Design Competition.

The Competition

In the 2026 SoutheastCon Circuit Design competition, the goal of the competition is to build an embedded control system that can control and monitor the water pump motor for the **Aquila Aerospace's** first rocket, the *Firebird-I*.

The competition goals are divided into different milestones that help organize the construction, test and documentation of the system. Points are given for each milestone completed, and additional points are available for useful work or additions beyond the minimum required for each milestone.

What are the project milestones?

The team at **Aquila Aerospace** have already assembled the motor, pump and plumbing and are eager to test the worthiness of those components before the first flight. To aid in their testing, they need two pre-flight milestones used for testing the new equipment.

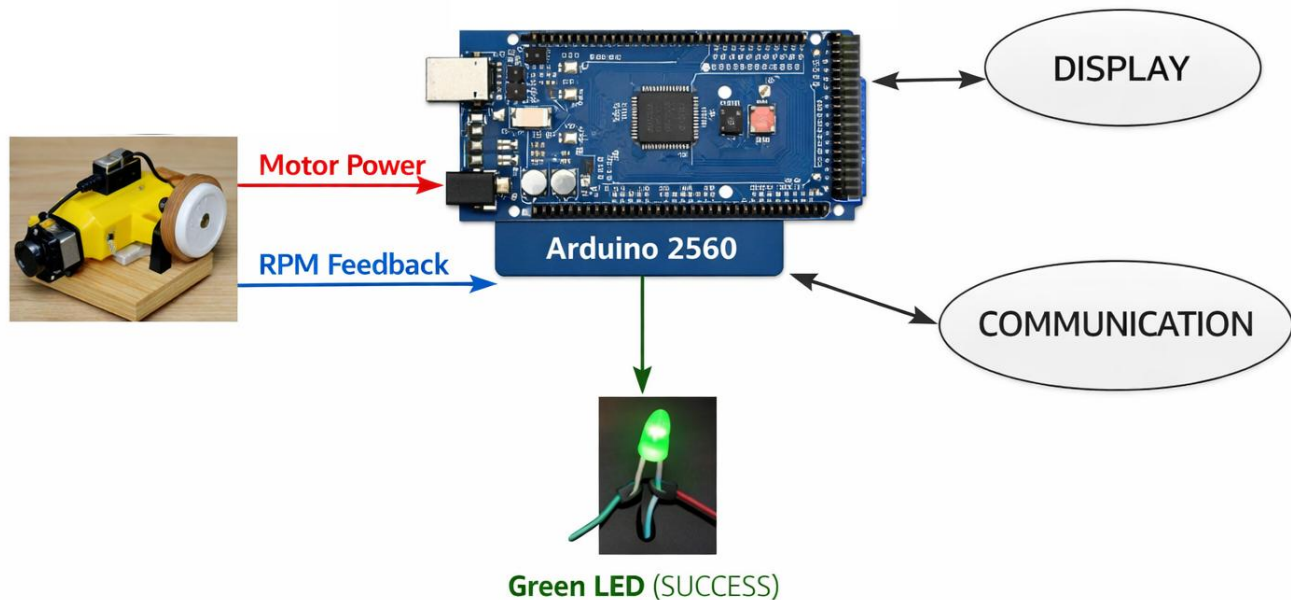
Milestone 1 (test jig) – In this milestone, the embedded controller should wait for 3 seconds after power on to energize the motor, and then monitor and display the motor's RPM. The display can be fairly simple for this milestone, including printing to the debug console, or use of the LCD and LED display devices.

The motor is rated at 200 RPM at full operating voltage of 6v, and the encoder disk is 20 slots per revolution, so the input signal from the IR based slot sensor is approximately 600hz, so interrupts are highly recommended for implementing this stage.

After powering on the motor, the system should give a visual feedback that the system is good (a green LED, or output to the LCD/LED screens, or the debug console).

This minimum functionality will allow the engineers to verify the motor is powered on and can ramp up to the required speed for the water deluge.

Completion of Milestone 1 is worth 200 points and should be documented with a short video of the operation. Note that since Milestone 2 includes all the features of Milestone 1, so teams can skip documenting this phase if they are confident they can reach milestone 2 (this saves some time at the risk of not getting as many points if Milestone 2 is not completed).



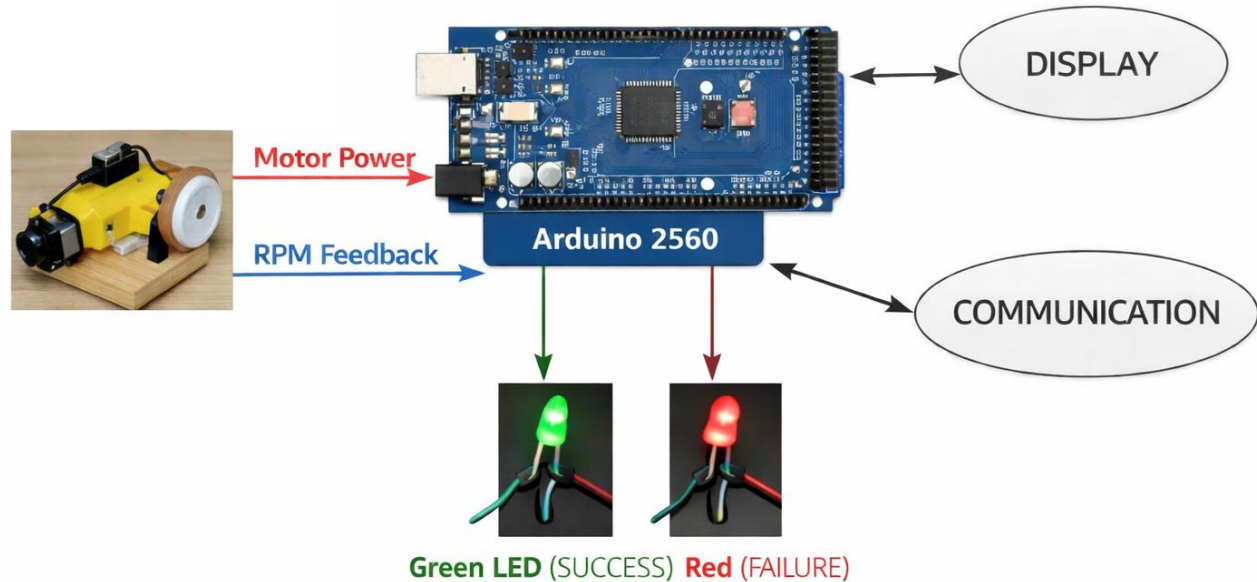
Milestone 2 (enhanced test jig) – This milestone builds on the previous milestone, but also adds in a few additional features, including visual countdown, slow start, and fault detection.

During the 3 second initial power on, the system should visually indicate the countdown timer on either the external LCD or LED display (not just the debug console). Once the three second countdown is completed, the motor should be slowly ramped up to final speed over a period of 2 seconds, displaying the speed during the ramp-up on the external LCD/LED display. If the motor is able to achieve the full speed within that time limit, a success visual indicator of a green LED should be turned on and kept on for the remainder of the system operation, unless a fault is detected.

If the motor cannot reach at least 75% of the full-on desired speed within the required 2 seconds, a visual indicator of a red LED should be turned on, and an optional audible alarm is also allowed. The system should continue to reach full-speed for an additional 5 seconds, and if successful, the fault should be cleared and the green LED should be turned on.

This additional milestone allows testing of the motor pump in a more real world scenario where problems could be detected and indicated.

Completion of Milestone 2 is worth 200 points and should be documented with a short video of the operation.



Milestone 3 (minimal viable product) – This milestone represents the completion of a first minimal viable product that can be used for launches and adds further fault detection and analysis.

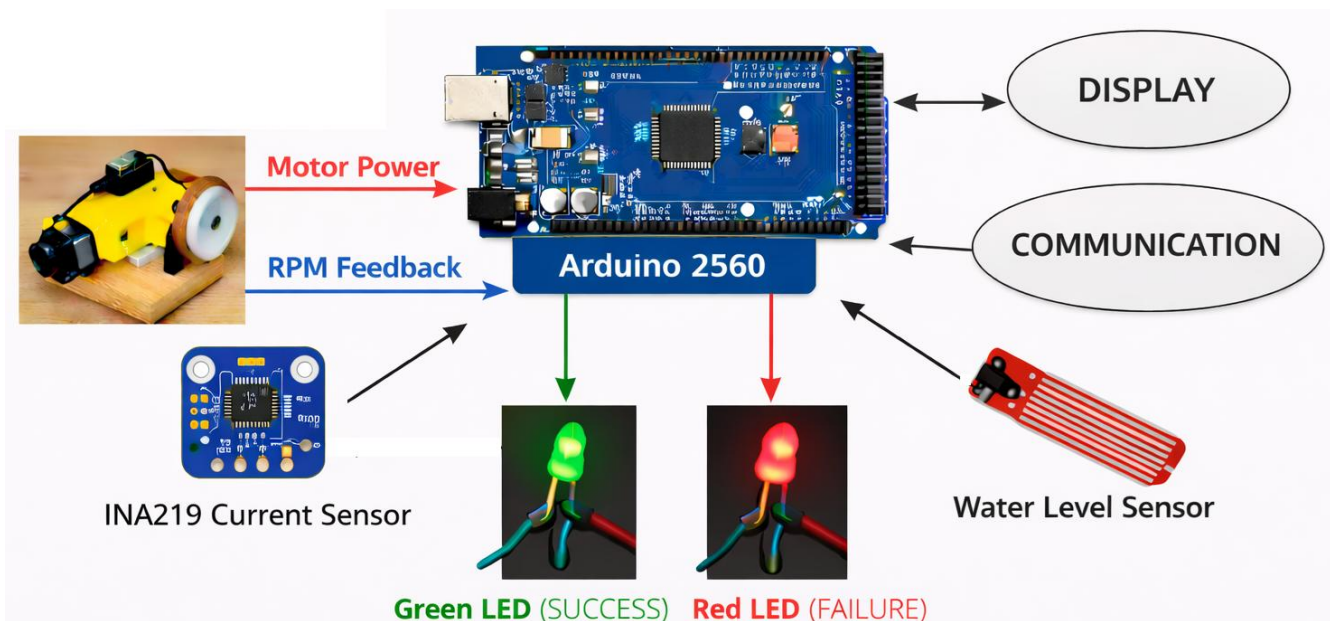
The power-on operation is the same as in the previous milestone with the exception of additional data collection and fault detection. The additional hardware includes a water level sensor and an INA219 current sensor.

The water level sensor should be checked to make sure that water is available before the motor is energized. If the water exists, the operation proceeds exactly as the previous milestone. If water does not exist, a fault condition should be indicated on the red LED and the external LCD/LED screen. The water level should also be sensed during the normal operation of the motor and a fault should occur immediately if water is not detected.

In addition to displaying the motor RPM, the motor's current should be measured with the INA219 and reported on the LCD/LED. Tests should be conducted to determine what is the normal current operation for the motor in previous runs and stored in the system. If the current rises 50% above the normal level, a fault indication should be indicated.

In either fault scenario, the motor should be immediately stopped, the red LED turned on, and the type of failure displayed on the LCD/LED.

Completion of this milestone is worth 300 points and should be documented with a short video of the operation.



Milestone 4 (your imagination) – This milestone allows your team to show your true skills in code and hardware development with more fault detection and analysis. In this milestone, use the included sensors and parts to add additional fault detection and analysis.

Here are a few ideas of what you could add to improve the system:

- ▶ Temperature / Humidity detection – use the DHT11 temperature and humidity sensor to detect a rise in temperature of the motor, or an increased humidity due to a possible leak in the pump. Note that the motor temperature does not rise enough to test this directly on the motor, so use an external heat source (your hand should be sufficient) to increase the temperature.
- ▶ Volatile gas – use one of the provided volatile gas sensors (two are provided in the two kits) to detect the rise of volatile gasses that could lead to a disastrous fire if it were to become in contact with the electrical motor.
- ▶ Fire – since there is no direct fire sensor, you can use the LDR (light dependent resistor) to indicate an increase in light, potentially due to a fire within the motor housing.
- ▶ Vibration detection – attach the accelerometer to the motor to detect and record vibrations and alert if the vibrations increase due to motor stress beyond a threshold you determine based on normal operation. A minimum display would be a fault red LED, and a status message displaying high vibration, or you could display the vibration on the Arduino built in analog plotter (accessed via Tools → Serial Plotter, YouTube has several good examples of this) or using an external Python / TK script similar to what was done in the AI homework / lab exercise.
- ▶ Audible noise – use the microphone (attached to the motor) to detect any variation in sound from the motor due to strain or stress on the motor.

- ▶ Visible system status – use the red, yellow, green stoplight LED array to show alarm (red), success (green), or in transit (yellow) while the motor is attempting to reach full speed, or to indicate any other near alarm indications.
- ▶ Look through the kits to see what other sensors and components you have and how they could be implemented – points are awarded for originality in the design.

This is an open-ended milestone worth 300 points, given at judge’s discretion. To achieve the most points within this milestone, teams must document what problem they were trying to solve or detect, what sensors or other parts were used, how the improvement was used to improve fault detection, and any testing and test results.

Competition Timeline

The competition will run at SoutheastCon 2026 on ‘lucky’ Friday, 13th, 2026 in Huntsville, AL.

- ▶ Teams are allowed early access to the competition room at 12:30, to allow time to set up.
- ▶ Rules are released at the start of competition at 1pm.
- ▶ Teams then have four (4) hours to design, build, program, and test and document their system.
- ▶ Teams must submit a report that includes their design of the hardware and software and any testing they completed, as well as any AI sessions submitted during the competition, and a short demonstration video of their system by 5pm. All of these deliverables are due before the close of the competition at 5pm – no time later, so be sure to allow time for this important step.
- ▶ Teams clean up their space, return items to the contest judges, and exit the room by 5:30pm.

Judging Criteria

Points will be awarded for:

- ▶ Completion of each milestone.
- ▶ Robust fault detection.
- ▶ Creative use of sensors.
- ▶ System reliability.
- ▶ Documentation and demonstration quality.
- ▶ Novel engineering approaches.

The scoring for each milestone is summarized below:

- ▶ 200 points for completing all the requirements of Milestone 1.
- ▶ 200 points for completing all the requirements of Milestone 2.
- ▶ 300 points for completing all the requirements of Milestone 3.
- ▶ 300 allocated to work done beyond Milestone 3 using the hints provided, or student’s own ideas.
- ▶ Partial credit for an incomplete milestone will be awarded at judge’s discretion, as well as extra points if a team goes beyond the minimum of a particular milestone.

There is a maximum of 1000 points for a team.

Deliverables

Teams must provide:

- ▶ A report describing the team's approach to the problem, how they tested the system, what sensors they used and how they used them to monitor the system and look for faults, a list of any issues they found and were able to resolve (or not resolve), and any future innovations or improvements if given more time or resources. This report should include a schematic diagram of the system, or at least a net-list of the connections between the Arduino, sensors, and actuators.
- ▶ Any code used in the final design must be submitted – this includes the Arduino code, and if implemented, any code running on a separate laptop for logging and display.
- ▶ Note that all use of AI must be documented by providing the complete chat showing the prompts given to the AI, and the AI response (how to do this may vary by AI used, but in ChatGPT, one easy way to do this is just typing <ctrl-p> to bring up the print menu, and then printing to a PDF file. It is recommended that teams make sure they start a new session at the beginning of the competition if there are any previous sessions so that only the contest specific area is reported.
- ▶ A short demonstration video of their working system.

Use of AI in the competition

This is the first time that SoutheastCon has allowed the use of AI in any of the student competitions. During the course of the competition, it is possible that changes or restrictions may have to be placed on all teams if that is warranted, but the hope is that will not need to occur.

Teams should write their own report and NOT use AI to generate the report. AI can be used to generate the graphics, schematic diagram, help format sensor data, etc, but teams are expected to be able to explain the system without the use of AI to show that they are aware of how the system is supposed to work.

It is also possible that at some point in the competition, AI may be disabled for all teams. If this happens, teams will be given a 15 minute warning to stop all AI and to capture any AI data.

Competition Spirit

Teams are encouraged to explore creative solutions and demonstrate engineering judgment. Artificial intelligence tools, documentation, and prior coursework may be used during the competition unless otherwise specified.

Teams are expected to uphold the IEEE Code of Conduct at all time.

What is provided to each team?

Each team receives the following Arduino and sensor sets for each station:

- ▶ [Elegoo MEGAL 2560 Arduino Kit \(tutorial here\)](#)
- ▶ [KEYESTUDIO 48-in-1 Sensors Modules Starter Kit for Arduino](#)

In addition, the following additional parts are available for the 2026 challenge:

- ▶ TT Motor Dual DC 3-6V Gearbox Motor
- ▶ IR Optical Interrupter Sensor
- ▶ Coded Disc Encoder (20 slots per revolution)
- ▶ INA219 I2C Bi-Directional DC Current Sensor

Teams also have access to additional items available, including:

- ▶ Tools, including wire cutters, scissors, pliers
- ▶ Miscellaneous discrete parts such as resistors and capacitors
- ▶ FNIRSI DSO-TC3 3-in-1 Handheld Oscilloscope (note there are not enough of these for all teams to have at the same time, so these are out on short loan and must be shared, or teams can use their own meters if available).

Hints and suggestions to each team

If you run into issues with the competition, here are a few helpful hints:

- ▶ Please remember you have full access to Google, YouTube, AI, or any online reference to aid you in looking for initial solutions and any followup issues you may encounter.
- ▶ Motors are inherently noisy – both audibly and especially electrically. If you run into problems with the noise interfering with your sensors, consider the following suggestions:
- ▶ Add capacitors to the motor, drivers, and across the breadboard power lines.
- ▶ The motor RPM is based on the full rated voltage of 6v, so the actual maximum RPM will be less, so do not use a hard-coded 200 RPM value.
- ▶ Remember to work as a TEAM – possibly dividing up the various tasks to different team members, writing the report, running AI chat sessions, wiring up and testing the hardware, developing the software, and testing the system – possibly rotating the job titles to different team members during the competition.
- ▶ Reach out to one of the contest judges – they are limited in what help they can provide, but could possibly offer some useful suggestions on where to look for solutions.
- ▶ Again, look to Google, YouTube, and AI for suggestions.

And a final message to the teams:

   Remember to have fun and good luck!   